In this brief statement, I will outline my activity versus expectation for each faculty pillar of teaching, research, and service.

Teaching Excellence

ERAU-WW COAS expects Associate Professors to demonstrate excellence in teaching. This is described in the Guidelines and Criteria document as follows:

> COAS embraces a philosophy of teaching effectiveness that emphasizes knowledge of subject matter, enthusiasm in the classroom, and continuous improvement through the incorporation of new teaching methods and materials. Faculty are expected to effectively communicate with students, facilitate learning in the classroom, and promote attainment of Student Learning Outcomes and General Education Competencies. These activities should be fostered through regular participation in pedagogical development opportunities and appropriate course maintenance and redevelopments.

I hold exceptional subject matter expertise in the courses I teach. As part of my Ph.D. in Mathematics and Statistics, I completed 33 graduate-level credit hours of mathematics courses [**CV, Notable Coursework, pg 13**], including Ph.D. qualifying exams in Matrix Analysis and Abstract Algebra. This high-level coursework allows me to contextualize undergraduate mathematics in a way that allows for productive abstraction and transfer to related topics of study. Abstract Algebra provides foundational mathematics for cryptography, programming, and physics while Matrix Analysis is critical to data analysis, machine learning, and 3D movement of objects (like a flying plane). For example, a common task in a Precalculus course is to solve a System of Equations. I can contextualize this task within aeronautics as 2-dimensional balance of forces (gravity, drag, thrust, etc), then use technology to show students how these calculations might be made by a computer for the "real" 3-dimensional version. For students more interested in machine learning, this same task of solving a System of Equations can be contextualized as computer image recognition. This allows me to spark student curiosity that can drive their engagement with mathematics.

As I learned completing 15 graduate-level credit hours focused specifically on teaching college mathematics [**CV, Notable Coursework, pg 13**], teaching well is more than just subject knowledge expertise. Research in Collegiate Mathematics (and Education in general) suggests subject knowledge acts in unison with pedagogical and social knowledge to deliver educational experiences. Thus it is critical to not only know the subject well, but also to know how to communicate these ideas at the level the student understands the concept. My continued research on how students develop mathematical ideas fuels my continued improvement as an instructor.

The union of my subject matter and pedagogical expertise allows me to provide exceptional teaching in ERAU-WW courses. This is evidenced by my overwhelmingly positive student evaluations [**Teaching Artifact: Student Course Evaluation Narrative**] that explicitly mention how my knowledge and enthusiasm create an effective learning environment. My research also affords me the opportunity to constantly incorporate new methods and materials. I have been part of three funded external grants from the National Science Foundation (NSF) Improving Undergraduate STEM Education (IUSE) program, received a mini-grant from a separate NSF IUSE grant, received an ERAU Faculty Innovative Research in Science and Technology (FIRST) grant, and received two ERAU-WW Faculty SEED grants that target improving specific Embry-Riddle courses [**CV, External/Internal Research Funding Experience, pgs 1-2**]. These research projects provide tangible improvements to student learning at Embry-Riddle, such as the Doenet mini-grant to develop autonomous discovery-based learning online [**CV, External Research Funding Experience, 8th item**]. I have also incorporated traditional active learning when teaching EagleVision sections, as noted by Zackery Reed's peer observation of my October 2022 Math 111 section. From his review:

> Dr. Chamberlain demonstrates a productive balance of direct instruction (i.e., lecture) and active learning activities. This suggests an intentional awareness of student attention span needs, and more importantly an implementation of currently known best practices for mathematics pedagogy.

Curriculum Development

I currently run two programs within the Department of Mathematics, Science, and Technology: the Minor in Applied Data Science and the Bachelors in Data Science. Both have seen early success in their enrollment numbers, as listed in the chart below. These programs leverage the world's interest in data science with the popularization of LLMs and other Artifical Intelligence software.

	Minor DSCI	Bachelor DSCI
August 2023	2	NA
January 2024	2	NA
August 2024	4	NA
January 2025	11	NA
August 2025	26	19
January 2026	42	34
August 2026	72	54

Given my extensive experience with course development during graduate school and while at the University of Florida [**CV, Teaching Experience, pgs 10-11**], I have completed several course developments, redevelopments, and updates. My developments have been widely praised by Instructional Design and Development (IDD) [**Teaching Artifact: IDD Support Letter**], faculty reviewers, and students. While my Summary of Teaching Effectiveness [**Summary of Teaching Effectiveness**] details all significant development efforts, I present the most impactful development below.

MATH 502 - Boundary Value Problems: I was asked to teach MATH 502 in August 2024 due to sustained student complaints and notices that students were dropping out of the Master's in Aerospace Engineering. Students were generally frustrated with the course structure and overly challenged by the material itself. I taught the course as it was in August 2024 and began making revisions as I taught it again in January 2025. I made several small changes to the course to address student complaints, including: (a) shifting all content back a week and creating a review of Ordinary Differential Equations for Module 1, (b) creating reflective discussion posts that encouraged students to explicitly link how the course topics related to one another, and (c) replacing synchronous oral exams with asynchronous oral exams. Creating the Module 1 review of Ordinary Differential Equations was enough to reduce student work on the proceeding Modules from an average of 30 hours per week to about 15 hours per week, putting it in line with expectations for a graduate-level course workload. Students generally praised the changes made to the course, with comments such as

> Great professor for the class, especially with this course because MAN this course was the hardest class I've had in years. Dr. Chamberlain should change ALL the videos to his style because HE can teach this like its NOTHING, such a great professor! (Aug 2024)
>
> The professor was really good at explaining the content when I had questions. I was impressed by how open and adaptive he was with feedback from the class. He recognized that somethings weren't working and changed quickly to ensure we learned in a better way. (Jan 2025)
>
> Out of the two courses I've taken this takes the cake. The lectures were well-structured, and Dr. Chamberlain's teaching style struck a great balance between theory and practical problem solving. I especially appreciated the timely, specific feedback on assignments. The course also made room for growth; policies and discussions encouraged iteration and improvement rather than one-and-done grading. The materials (notes, example codes, and references) were organized and directly useful for quizzes and projects. (Aug 2025)

Scholarly and Professional Activity

My scholarly activity, compared to total expectations outlined in the COAS ERAU-W Academic Guidelines and Criteria for Tenure and Promotion to Associate Professor, is provided in Figure 1.

	Peer-Reviewed Publications			Scholarly Presentations					Funding Proposals			
	Book Chapters	Journal Articles	Proceedings Papers	International Conference	National Conference	Regional Conference	Local Conference	Invited Talks	Funded External	Unfunded External	Funded Internal	Unfunded Internal
Total	**1**	**7**	**9**	**3**	**19**	**1**	**4**	**6**	**4**	**6**	**4**	**0**
Year 1: AY21-22		1	2		4	1		1	1		1	
Year 2: AY22-23		2	1		1		1	2	1		1	
Year 3: AY23-24			1		4			2	1	1	1	
Year 4: AY24-25	1	1	4	1	5		2	1		2	1	
Year 5: AY25-26		3	1	2	2		1		1	3		
Year 6: AY26-27					3							
Actual Sum Total	17				23		4	6	14			
Expectation	5				5		NA	NA	1			

Figure 1: Summary table of activity for AY1-6

Of the peer-reviewed publications, 10 of 17 were first-author publications and 3 of 17 included student researchers. Publications were made in high-quality venues focused on education. The book chapter was published in *Teaching and Learning Mathematics Online 2e*, the second edition of a series focused on presenting meaningful and practical solutions for teaching mathematics online. Proceedings papers were primarily published in the proceedings of the annual conferences on Research in Undergraduate Mathematics (6 of 9) and American Society for Engineering Education (2 of 9). Journal articles went to high-quality journals such as *Journal of Mathematical Behavior*, *Active Learning in Higher Education*, *Intersection: A Journal at the Intersection of Assessment and Learning*, and *Problems, Resources, and Issues in Mathematics Undergraduate Studies*.

All proceedings papers had an associated international (1) or national (8) presentation. The other 2 international presentations were associated with an internal grant (FIRST) and included 1 graduate and 2 undergraduate student researchers. The graduate student decided to continue with their PhD after finishing their involvement with the project, stating in their exit interview that working on the project helped them learn how to structure code for a large-scale research project and formalize it for publication. Other national presentaions were primarily at the annual Mathematical Association of America MathFest conference (5 of 11).

In terms of external grants, I have been a co-PI on (3) funded NSF grants and PI on (1) subaward attached to NSF grant while being PI (3) and co-PI (3) on unfunded external grant proposals. The chart does not include a resubmission of an unfunded but competitively-rated grant in October 2026 nor does it include a projected resubmission of another unfunded but competitively-rated grant in January 2027. I have had stellar success with internal funding: I have been PI (2) and co-PI (2) on funded college-level (3) and university-level (1) internal grants. My summary of scholarly activity provides a short description of how my active research projects contribute to my overarching research agenda on utilizing machine learning and hypothetical learning trajectories to model student thinking and learning [**Summary of Scholarly Activity**].

Service

Finally, ERAU-WW COAS expects Associate Professors to provide service both to ERAU-WW and beyond. Specifically, the Guidelines and Criteria document call for:

> Appropriate service to their department, college, campus, university, and professional community. Service commitments for a candidate will be determined annually in collaboration with the Department Chair. Candidates' portfolio should include evidence of varied, sustained and, typically, progressive levels of service.

I have had significant service to the department and college, as summarized in my [**Summary of Service**]. Of these, one of my most impactful services was stepping in as Associate Department Chair after former department chair Karen Keene suddenly passed in February 2023. In my annual evaluation, interim department chair Beverly Wood stated *"He has been a key ingredient to the recovery of MST following the sudden death of*

Karen. Not only has he helped to maintain the department but has added efficiencies that benefit us all". One of these efficiencies was the Faculty Activity reporting survey and database I created, first for the department and then on behalf of the entire college. This system allowed for ease of faculty submitting their activity to a single location and automated, formatted reporting of faculty activity for efficient monthly and bi-yearly reporting. I met with representatives of both College of Aviation (Kimberly Luthi) and College of Business (Maria Petrescu) to discuss how they could implement a similar system, and the new Watermark activity reporting was partially modeled after my Faculty Reporting system. In addition to serving as Associate Department Chair from February 2023 to May 2025, I was also Acting Department Chair from January 2024 to May 2024 as the college continued to search for a permanent Department Chair. My leadership during this time was both publicly and privately praised, such as John Griffith (former MST chair) stating:

> We have talked before that you would be a great department chair should you want to do that in the future. You really took care of the department in your roles as acting chair and associate chair. You have a bright future. Thanks for being so great to work with!

Beyond the university, I have been an active leader in my overarching professional community, Mathematical Association of America (MAA), as well as the smaller research community Research in Undergraduate Mathematics Education (RUME). As part of the MAA, I have chaired the subcommittee on Technology in Undergraduate Education, served on the MAA Council for Teaching and Learning, and served on the committee for Teaching of Undergraduate Mathematics [**MAA Service Letter**]. As part of RUME, I have served on committees dedicated to leadership nomination and annual conference programming as well as run for leadership positions (Treasurer and Secretary) [**Request to Run for RUME Leadership**]. Beyond my professional communities, I have served several times as a National Science Foundation Grant Reviewer and am a frequent journal reviewer, sought out for my expertise in APOS Theory, proof, and structural equation modeling.

Promise for Sustained Future Development and Scholarly Activity

I plan to continue developing along my three major lines of inquiry outlined in my [**Summary of Scholarly Activity**]: Generative AI in Education, Educational Activities in Online STEM Education, and Data Analysis for Engineering Education.

Regarding work on Generative AI in Education, I have an under review paper that utilizes the automated diagnostic item generation from my prior work diagnosing student misconceptions to assess the efficacy of Generative AI diagnosing potential student thinking. I am also lead editor of a special issue on Generative AI in the Teaching and Learning of Mathematics with *Problems, Resources, and Issues in Mathematics Undergraduate Studies* that is on track to be published by end of year 2027. These efforts are in addition to the 2026-2027 NSF grant *Convening to Investigate the Impact of AI-Powered Pedagogies in Postsecondary Mathematics Education* I am co-PI on alongside MAA's Chief Strategy & Operations Officer Audrey Malagon. These efforts illustrate my ability to weave teaching, scholarly activity, and service together in a manner consistent with my research agenda for productive future activity.

Regarding work on Educational Activities in Online STEM Education, I will continue to apply to NSF IUSE grants in collaboration with Georgia State University to develop diagnostic calculus assessments. Grant submissions in both 2024 and 2025 were reviewed as "Competitive" but were not funding, suggesting the project is meritorious and provides a strong foundation for continued refinement and resubmission [**Diagnostic Calculus NSF Reviews**]. Note Georgia State University is an R1 university and this collaborative effort would further bolster Embry-Riddle's continuing climb as a research university.

Regarding work on Data Analysis for Engineering Education, I will continue to support research efforts hosted by the ERAU-WW College of Aviation. This allows me to apply my educational data science research expertise directly into ERAU signature research fields. My work in the NSF-funded Research Scholars Program led by Robert Deters (COA) contributed to my inclusion on the under review NSF IUSE proposal led by Ghazal Barai (COA) titled *Appraising Algorithmic Evidence: How AI-Generated Workforce Signals Shape STEM Learners' Motivation and Faculty Reliance on AI.*

Summary of Student Evaluations

Student evaluations represent one component of my evidence of teaching effectiveness. I interpret them alongside peer observations, curriculum-review feedback, evidence of student learning, and reflective evaluation of course design. Students have responded overwhelmingly positively to my methods of teaching courses at ERAU-WW. As noted in my [**Reflection on Student Course Evaluations**], 86/92 were entirely positive and noted the extensive feedback I provide students, the numerous helpful resources I have when needed, and the accepting environment I engender with my process over product mentality. A summary of student evaluations by term, compared to college instructor evaluation average, are provided in a table below.

			# Students	# Respond	% Respond	Instructor Experience Mean	College IE Mean	Positive Comments to Instructor	Neutral Comments to Instructor	Negative Comments to Instructor
Term	**Year**	**Course**	**421**	**268**	**64%**	**4.67**	**NA**	**86**	**6**	**0**
September	2021	MATH 143	17	12	71%	4.48	4.55	3	1	
October	2021	MATH 111	10	9	90%	4.81	NA	4		
October	2021	MATH 143	25	19	76%	4.59	NA	6		
January	2022	MATH 111	29	16	55%	4.43	4.52	2		
January	2022	MATH 111	32	20	63%	4.61	4.52	6		
August	2022	MATH 106	34	25	74%	4.67	4.49	8		
October	2022	MATH 111	20	14	70%	4.91	4.75	4		
January	2023	MATH 111	34	19	56%	4.61	4.46	5	1	
March	2023	MATH 111	35	21	60%	4.75	4.46	7		
August	2023	MATH 111	24	16	67%	4.84	4.52	7		
October	2023	MATH 111	29	14	48%	4.63	4.43	4		
January	2024	CSCI 251	4	4	100%	4.72	4.53	2		
August	2024	CSCI 251	6	3	50%	4.15	4.56	2		
August	2024	MATH 502	17	8	47%	4.78	4.56	5	1	
January	2025	MATH 502	20	16	80%	4.62	4.55	5	1	
August	2025	STAT 412	15	8	53%	4.74	4.67	1		
August	2025	MATH 502	11	8	73%	4.45	4.67	5	2	
October	2025	DSCI 201	10	5	50%	4.8	4.67	5		
October	2025	STAT 412	15	8	53%	4.75	4.67	1		
January	2026	CSCI 251	9	6	67%	5	4.65	3		
January	2026	MATH 502	25	17	68%	4.71	4.65	1		

Curriculum Development

Additional evidence of my effectiveness as an instructor is in the evaluation of curriculum I have developed. When creating courses, I create all homework assignments (no third-party homework platforms), write summarized notes of open-source textbooks, and ensure students have a balance of assignments that target conceptual, technical, and metacognitive development. While a full list of developments can be found in the Watermark [**Teaching/Mentoring**] report, I summarize three impactful developments below.

CSCI 251 - Introduction to Programming for Data Science: This new course, initially built for the Minor in Applied Data Science and later incorporated into the Bachelors in Data Science, teaches students how to program in order to manage and analyze large data sets. The courses does not use any paid third party platform – all learning resources and assignments were created for the ERAU-WW course specifically. Some of these are traditional assignments, such as open-response Canvas quizzes, while others encourage active groupwork, such as groups developing documentation specific to the course. This allows students to get genuine feedback before they write entire programs on their own. Students universally praised the course, with comments such as:

> The course was fantastic, the instructor's assignments were consistent and understandable, challenging but never overwhelming. (Jan 2026)

> All of the different teaching methods were really helpful... What helped most of all was being able to get my hands on the program itself and experiment with it and just have fun with it. (Jan 2026)

As one of the first courses in both the minor and bachelors' programs, it is important to have an engaging and interesting course to intice students to continue their program.

DSCI 201 - Introduction to Applied Data Science: Another instructor who is no longer at ERAU created DSCI 201 but did not teach it before leaving. The course was rife with low-level engagement assignments that only required students to write a couple sentences based on something they read. I began creating engaging assignments that utilized students' coding skills to implement machine learning algorithms. Comparative examples of assignments are provided in [**Optional Teaching Artifact - DSCI 201 Comparison**]. Students praised my progressive edits to the course with comments such as:

> I found the course well-paced and interesting. The instructor was engaging and encouraging.
>
> I would recommend that the videos Dr. Chamberlain made for CSCI 251 be used for this course, as I found them especially helpful and easy to follow along with.
>
> The Jupyter notebooks along with specific examples online and linking to the documentation helped me see the learning objectives clearly.

Math 111 - Pre-Calculus for Aviation: In the first update, I rewrote the majority of discussions to improve students' metaskills as well as encourage students to engage with content beyond drilling skills. By metaskills, I mean students' ability to reflect on their study skills and evaluate mathematical work. Early discussions were designed to improve how students study for mathematics: both in study habits and with actually creating study materials. Later discussions focus on difficult mathematical concepts such as vector movement. One discussion in particular has received both hate and love by adjuncts. The question requires students to reason how to compensate for wind when flying between two cities. The question is written so that there are numerous ways students can reason through and "solve" for the specific correction in the problem. This led to instructor comments such as:

> I have a feeling that this problem could be too complicated in an introductory level course. Admittedly, even I had a hard time figuring out how to tackle this problem until I looked at the solution guide. I have never seen such an applied vector problem before this one.
>
> Attached are three students approach to the problem. The Answer Key provided shows a different approach to working the problem. Could you look over these students approach and provide me with any suggestions on how to best help them?
>
> Several of my students have asked for the velocity or speed of the airplane. I told them they have all the information that is needed to solve the problem and suggested that they make a sketch. I am afraid this isn't going to go too well!
>
> The new Module 7 Group Discussion Exercise in Math 111 was a tremendous addition to the course. Cognitive presence was demonstrated significantly in the various ways students sketched, formulated, solved and analyzed the given problem. Several students supplemented the work of others through extensive collaboration and tackling the problem from different angles... literally and figuratively! This was a welcome addition to an "aviation" course which up until recently didn't have many aviation applications or examples.

All of these comments point to an increased cognitive demand for the discussion post. Many students were and continue to be able to reason through the discussion post by the end of the week. This suggests the question is not too complicated, but IS a spike in difficulty from what they might see on the homework and exams. The second comment in particular highlights how there is no one way to reason through the task, which was precisely the point. The third comment points to the need for instructors to engage with the task beyond "correct" or "incorrect" and thus provides a genuine opportunity for teacher-to-student learning.

Overview of Research Agenda

As a researcher, my interests revolve around utilizing technology to improve undergraduate education. Before starting at ERAU, I began using programming languages to build automated student-to-expert interactions that are critical to improving student learning. I have since expanded my computer skills to include machine learning through a 5-course Coursera specialization. I am now uniquely positioned as a multi-disciplinary expert in mathematics, mathematics education, and data science.

At the macro learning-level, I study how different types of educational activities (e.g., discussions, quizzes, exploratory activities, digital manipulatives) may impact student understanding. At the micro learning-level, I create models that predict how students may respond to individual questions that are then used to provide automated feedback. My wide range of educational research, statistics, and programming skills have also led to productive collaborations where I act as an expert in data collection and qualitative/quantitative data analysis to improve undergraduate learning. My active research projects reflect these varied data science skills that all revolve around improving online education for college students.

Given the numerous successful funding proposals I have been a part of, associated with different research projects and collaborators, I have organized my research experiences by topic and summarized how they fit within my research agenda.

Generative AI in Education

Generative AI became a massive disruption to education, especially online education, in early 2023. I have since become increasingly involved in research on Generative AI's impact in education.

Generative AI Discussion Feedback: In Fall 2023, Alex Rister organized a cohort of faculty from various COAS-WW departments to investigate the extent to which Generative AI could provide feedback to student discussion responses. The team was awarded an internal grant and collected data from several courses. As part of this project, I contributed to 1 national conference presentation, 1 local conference presentation, and 2 accepted journal articles.

Generative AI in Writing Tasks: In Fall 2024, I began working with Emily Dux Speltz on Generative AI use in both academic and professional writing. We received an ERAU FIRST grant in Spring 2025 through which we have interviewed professionals and students after first recording their use of Generative AI to complete a professional or educational writing task. As part of this project, I have contributed to coordinating the 3 student research assistants on the grant and 2 international conference presentations. My prior research with Meghan Velez (University of Central Florida) has also led to a cross-institution NSF grant submission between UCF, ERAU, and USF to conduct a longitudinal study on student use of Generative AI in writing assignments, though it was not funded at this time.

Generative AI in Mathematics Education: As part of my responsibilities as chair of the Mathematical Association of America (MAA) Subcommittee on Technology in Mathematics Education, I have advised various MAA committees on the potential impacts of Generative AI in mathematics education. This led to leading a submission of an external Spencer Grant with MAA that would create an invite-only conference to discuss and spearhead research on Generative AI in Mathematics Education. After this Spencer Grant was declined, it was subsequently submitted to NSF and awarded in May 2026. The conference took place in June 2026 and produced a statement on Generative AI in Mathematics Education that has been approved by the MAA Board of Directors: https://maa.org/resource/statement-on-ai-in-higher-education/.

Educational Activities in Online STEM Education

While Generative AI has tasks it can do well (summarizing materials, translating languages, and small-scale coding tasks), it is not yet capable of providing nuanced feedback to student work without error. Thus I have been exploring alternative machine learning methods to detect patterns in student thinking and provide automated feedback. This has led to two threads of related activity: creating autonomous educational assessments for student learning and autonomously evaluating assessments for instructor review.

Autonomous Feedback to Students: Before joining ERAU-WW, I created a mastery-based College Algebra course at the University of Florida. Unlike a traditional course where content is presented on a schedule regardless of a student's performance, mastery-based courses have students focus on a small number of learning objectives until they display mastery, then allow the student to continue on to new learning objectives. In Fall 2023 I received a small external grant to implement a new educational technology (DOENET) to create a dynamic activity for inclusion in ERAU-WW's College Algebra course. Like my prior work at UF, I identified patterns of student thinking for inclusion in a College Algebra word problem. However, I was able to leverage the dynamic capabilities of DOENET to display and hide content on the webpage based on how a student responded. This dynamic activity was thus able to act as a "Choose Your Own Adventure" where students would follow paths of thinking, correct or incorrect, until it prompted them to return to some critical juncture with additional feedback on how they could change their thinking. In collaboration with colleagues at Georgia State University (GSU), we submitted a Collaborative Proposal to NSF in Fall 2024 and Fall 2025 to develop these types of dynamic activities as well as the autonomous feedback homework type assignments for GSU's Calculus 1 courses. The proposal was reviewed as Competitive each time but was not funded. This work has produced 1 journal article under review, 1 published journal article, 2 conference proceedings papers, 6 national conference presentations, and 2 local conference presentations.

Conceptual Understanding in Asynchronous Discussions: In Spring 2023 I led another internal SEED grant with Karen Keene and Zachary Reed to investigate how students' conceptual understanding progressed in asynchronous discussions in Calculus. We eventually published a methodology for identifying conceptual understanding and its progression in asynchronous discussions. This collaborative project was cut short after the untimely passing of Karen Keene in February 2023. These two SEED grants produced 1 book chapter, 2 conference proceedings papers, 2 national conference presentations, 1 regional conference presentation, and 1 local conference presentation.

Autonomous Feedback to Instructors: My efforts with autonomous feedback to instructors led to Emily Faulconer and Beverly Wood asking me to join their NSF grant when I started at ERAU in Fall 2021. In this project I worked with undergraduate student research assistants to qualitatively code student cognitive and social moves in online discussions. We then began extracting features from student discussion posts that would allow for autonomous qualitative coding and thus autonomous reports to instructors to provide top-level summaries of student participation in discussions. As Generative AI was quite proficient at summarizing by the end of 2024, the goal of automating qualtiative coding became moot. The project did provide insights on students' cognitive and social moves in a course though and was productive, producing 2 journal articles, 1 conference proceeding, 2 national conference presentations, 1 invited talk, 3 white papers, and 2 editorials.

Data Analysis for Engineering Education Projects

Since my expertise in data science can be applied to various fields outside of my immediate research interests, I continue to provide data science support for research projects led by ERAU-WW Engineering faculty. This allows me to support ERAU's signature research fields of aviation and aerospace education with my expertise in educational data science.

Research Scholars Program: In Spring 2023, I was approached by engineering faculty Robert Deters and Brent Terwilliger to act as the data scientist on their continuing submission of the Research Scholars Program. This project focuses on developing resources for supporting undergraduate research in online programs. This second stage expanded the impact to all ERAU-WW students, helped develop a cross-campus Undergraduate Research minor, expanded on resources to support undergraduate research, and studied the effects of these efforts. This project produced 1 journal article under review, 3 conference proceedings, 5 national conference presentations, 1 white paper, and 24 workshops.

Autonomous Curriculum Suggestions: In Fall 2025, I was approached by engineering faculty Ghazal Barari and Robert Deters to utilize artificial intelligence and machine learning to support curriculum updates based on curriculum signals from the workforce, such as skills detailed in job calls. This NSF proposal was submitted in November 2025 and deemed Competitive but not funded. It has since been revised and resubmitted to another NSF program in October 2026.

ERAU Service

A full list of the varied service activity I've performed over the past five years can be found in the automated Watermark [**Service**] report. This letter synthesizes that activity to illustrate sustained and progressive levels of service among 3 threads: college and university service, departmental service, and professional society service.

College and University Service

Given my extensive college and departmental service at the University of Florida (**[CV, pg 12]**), I immediately sought out impactful service opportunities at ERAU. In my first year I was elected the ERAU-WW Faculty Senate Academic Technology committee. With my expertise in academic technology, I was voted chair of the committee for all three years. Primary duties for the committee included systematic review of various educational technologies such as YellowDig. This trans-college service allowed me to build collegial relationships with faculty outside of my college such as with Brent Terwilliger (COA) who I eventually submitted an NSF grant with. It also allowed me to directly interact with university initiatives as an expert in educational technology, such as participating in the hiring of a Director of IT Service Management & Worldwide IT Service as well as the ERAU-WW End-of-Course Faculty Survey committee to focus student evaluation questions. My work on the council was praised by the Vice Speaker of the Worldwide Faculty Senate [**FS Committee Chair Evaluation**]. I then waited for another committee position to open related to my expertise, which occurred in October 2026 when I was elected to the ERAU-WW Faculty Senate Undergraduate Curriculum & Degrees committee. Throughout this time I also served as a university reviewer for internal grants and was frequently called upon to review mathematics and engineering-adjacent research grants.

At the college level, I served on the Faculty Council from 2022-2023, leaving after accepting the Associate Department Chair role in February 2023. During my short tenure on Faculty Council, I assisted Alex Rister in initiating a Faculty Council Collegiality Award to address faculty concerns on receiving a collegiality award from administration. My short time on the Faculty Council was praised by its chair [**Faculty Council Member Evaluation**]. I've also served the college in ad hoc forms. I volunteered to lead the Fall 2025 efforts to revise the Faculty Council charter after faculty voted to temporarily suspend the council. These efforts led to a successful restart of Faculty Council in January 2026. In 2024, I served as an inter-department member to hire a tenured Department Chair for the Department of Human Factors and Social Science. And from August 2023 until July 2026 I created and maintained a Faculty Activity Reporting system on behalf of the college. This system allowed for ease of faculty submitting their activity to a single location and automated, formatted reporting of faculty activity for efficient monthly and bi-yearly reporting. I met with representatives of both College of Aviation (Kimberly Luthi) and College of Business (Maria Petrescu) to discuss how they could implement a similar system. Maria thought the set up was excellent, stating in private email correspondence:

> I love this! Your survey is so much better than the Forms we've been using so far. If you don't mind exporting the survey so I can import it into my Qualtrics, as well as your Python script, that would be great. Our senior associate dean who's been handling this left, and I'm trying to catch up on this and make it more efficient.

Maria was involved in creating the new Watermark activity reporting and confirmed it was partially modeled after the Faculty Reporting system I shared with her.

Department-Level Service

At the department level, I have sustained high-level leadership since accepting the Associate Department Chair role in February 2023. As Associate Department Chair, I performed all the routine administrative duties as well as used my expertise in data science to innovate admin processes, creating significant efficiencies. For example, course monitors in the department asked for lists of instructors teaching courses they were monitors for by term and I created programming scripts to create such lists from raw Campus Solutions

reports. In addition, course monitors asked for a yearly accumulation of end-of-course evaluations for courses they monitored, which I created programming scripts to provide.

After the department chair hiring failed in January 2024, I accepted being Acting Department Chair (while still being Associate Department Chair) until another hiring cycle could be completed successfully. I stayed on as Associate Department Chair for another year to assist the newly hired chair, Tim Smith, and passed on the role in order to start as the Program Chair for the new Bachelor in Data Science in August 2025. The first year of the program saw major success and had 54 declared students by August 2026. During these years of administrative service I continually shifted the courses I was mentor for as the Department of Mathematics, Science, and Technology had heavy faculty turnover. I also chaired one tenure-track candidate search in Physical Science in 2024-2025 while participating as a member in three other candidate searches: tenured candidate in Computer/Data Science in 2026, tenured department chair in 2024 [**Higgins Search Letter**], and tenure-track candidate in Data Science in 2022-2023.

Professional Society Service

At the professional society level, I have also sustained high-level leadership in the Mathematical Association of America (MAA). From 2021 until 2022 I was a member of the MAA Subcommittee on Technologies in Mathematics Education (STME), then took over as STME chair and have led the committee since. STME regularly hosts contributed paper sessions at conferences, produces virtual programming surrounding innovative technology in mathematics education, has hosted a 2-day virtual conference on Generative AI in mathematics education, and led multiple grants in collaboration with MAA to support mathematics instructor use of Generative AI. My leadership in STME has been praised, with the chair of the Committee on Teaching in Undergraduate Mathematics (CTUM) and chair of the Council on Teaching and Learning (CTL) inviting me to chair CTUM starting June 2028 [**CTUM Chair Invitation**].

Beyond my formal STME involvement with MAA, I have also been informally included in MAA initiatives and events. In February 2025 I was invited to an informal meet-and-greet with a representative at the Gates Foundation as part of the MAA's work in connecting the Gates Foundation with exemplary MAA members [**Informal Gates Meet and Greet**]. I was also invited to represent the MAA to talk about Generative AI in mathematics education at a *Coalition for National Science Funding* (CNSF) that would include scheduled meetings with congress members [**CNSF Invitation**], though the MAA later decided not to attend the event. To capture the exceptional formal and informal service I've performed for MAA, I requested and attached a letter of service from MAA's Chief Strategy and Operations Officer Audrey Malagon [**MAA Executive - Service Review Letter**].

Outside of my MAA service, I am a well-respected member of the Research on Undergraduate Mathematics Education (RUME) community. I frequently present at the annual conference on Research in Undergraduate Mathematics Education and have led a yearly workshop on Technology in Mathematics Education. Given my profile in the community, I am frequently requested to run for RUME executive positions [**Email Request RUME Executive Committee**]. My work in RUME and service in MAA have also led to a wider profile as an expert in education. For example, I cannot share the entire invitation to serve on a particular NSF review panel as it contains confidential information, but I was invited by an NSF program officer who said:

> Dear Dr. Chamberlain, We've not yet had the opportunity to meet, but I've been aware of your work from my role as a program officer in the STEM Education Directorate at the National Science Foundation.

MAA's headquarters in Washington DC shared a building with the National Science Foundation offices for a few years between the sale of the original MAA headquarters and their relocation to a new permanent building. This short time allowed for excellent relationships to be built between the two organizations and STEM Education Program Directors frequently attend MAA conferences and events. My service in MAA has allowed me to build relationships with these NSF Program Directors to further raise my profile in the mathematics education community.